

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Lesson 4: Classification and Evaluation Metrics

Technologies for Artificial Intelligence

Fabio Antonini, Università degli Studi dell'Aquila

30 October 2026

Before we start

- Exercise 3, which we discuss now
- Ridge, Lasso, and the choice of the penalty strength

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Today: from a number to a decision

- Logistic regression: the model and why the cost had to change
- The confusion matrix, and why accuracy lies
- Precision, recall, F1, and the threshold nobody chose
- Receiver operating characteristic (ROC) curves, and where they mislead
- Choosing a threshold from money

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The question has changed

- Lesson 3: given a house, **how many euros?**
- Lesson 4: given a disk drive, **will it fail?**
- Answer with a probability: $0 \leq P(y = 1 \mid x) \leq 1$
- $y = 1$ if the drive failed, $y = 0$ if not; x holds its six counters

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The fleet

- **8,000 disk drives** in a data centre
- Six SMART counters each: Self-Monitoring, Analysis and Reporting Technology
- Reallocated sectors, spin retries, read and seek error rates, temperature, hours in service
- Label: did it fail within thirty days?
- `train_test_split(..., stratify=y)`: 6,000 to train, 2,000 to test

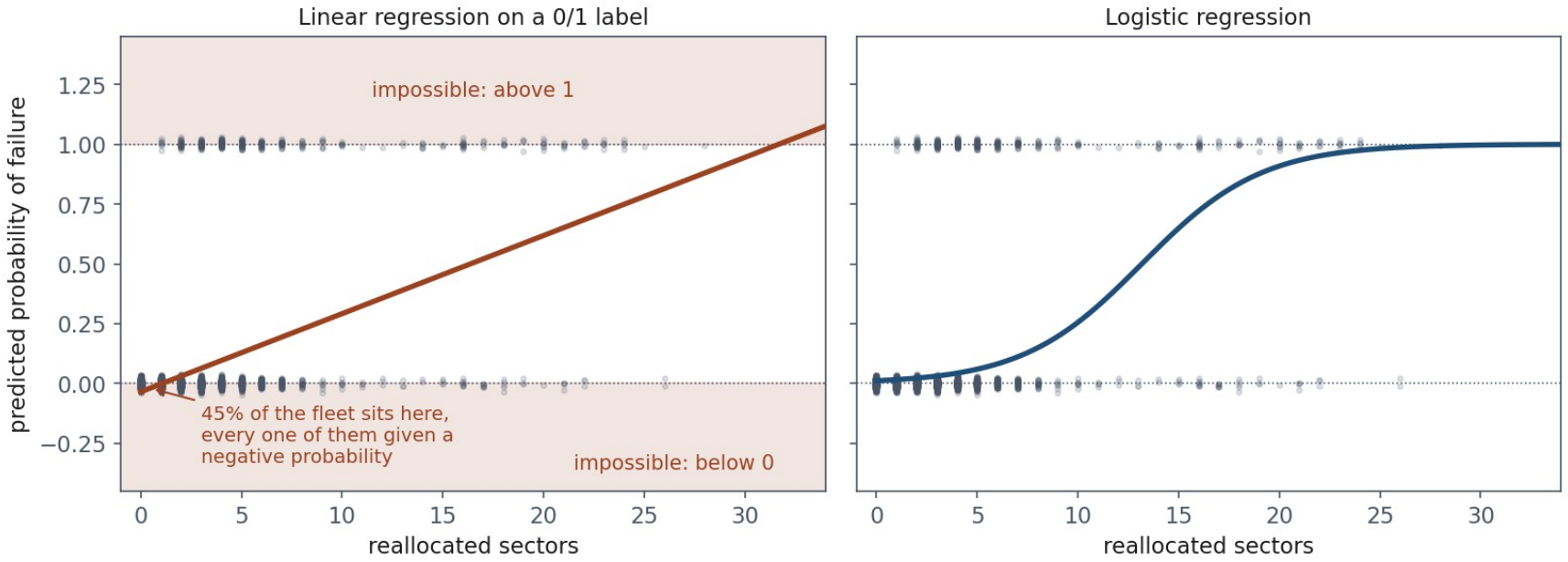
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

306

- **306 of the 8,000 drives failed**: 306 ÷ 8,000 = 3.8% of the fleet
- Which means 100% − 3.8% = 96.2% did not

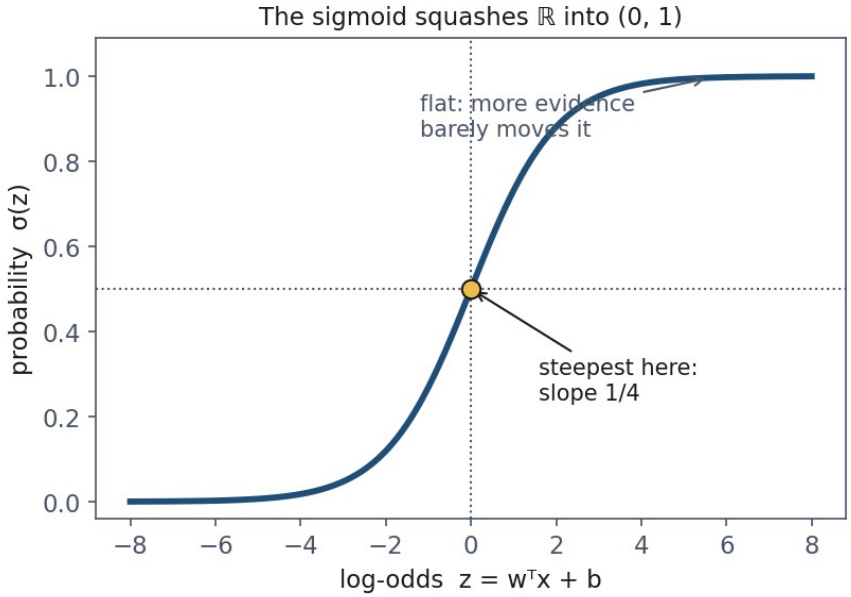
UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

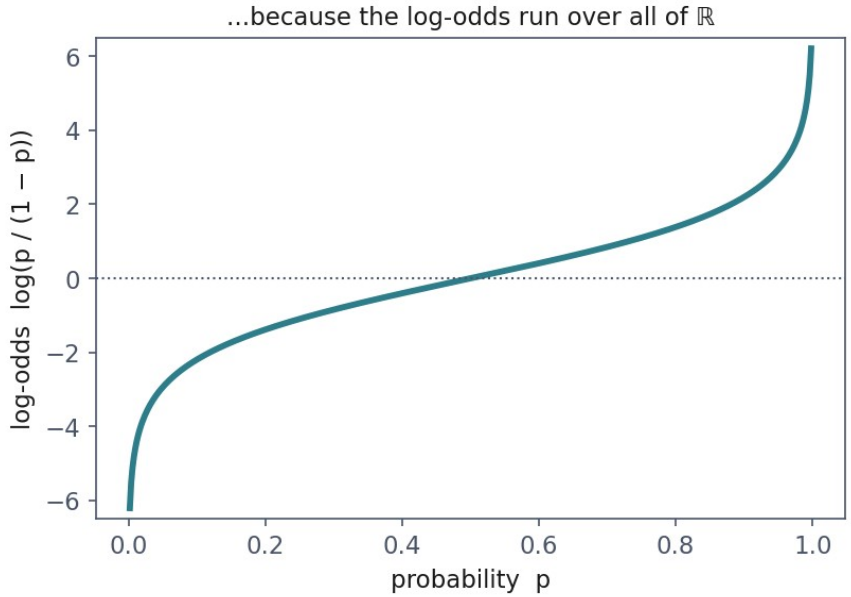
What goes wrong if we just fit a line



UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

The sigmoid





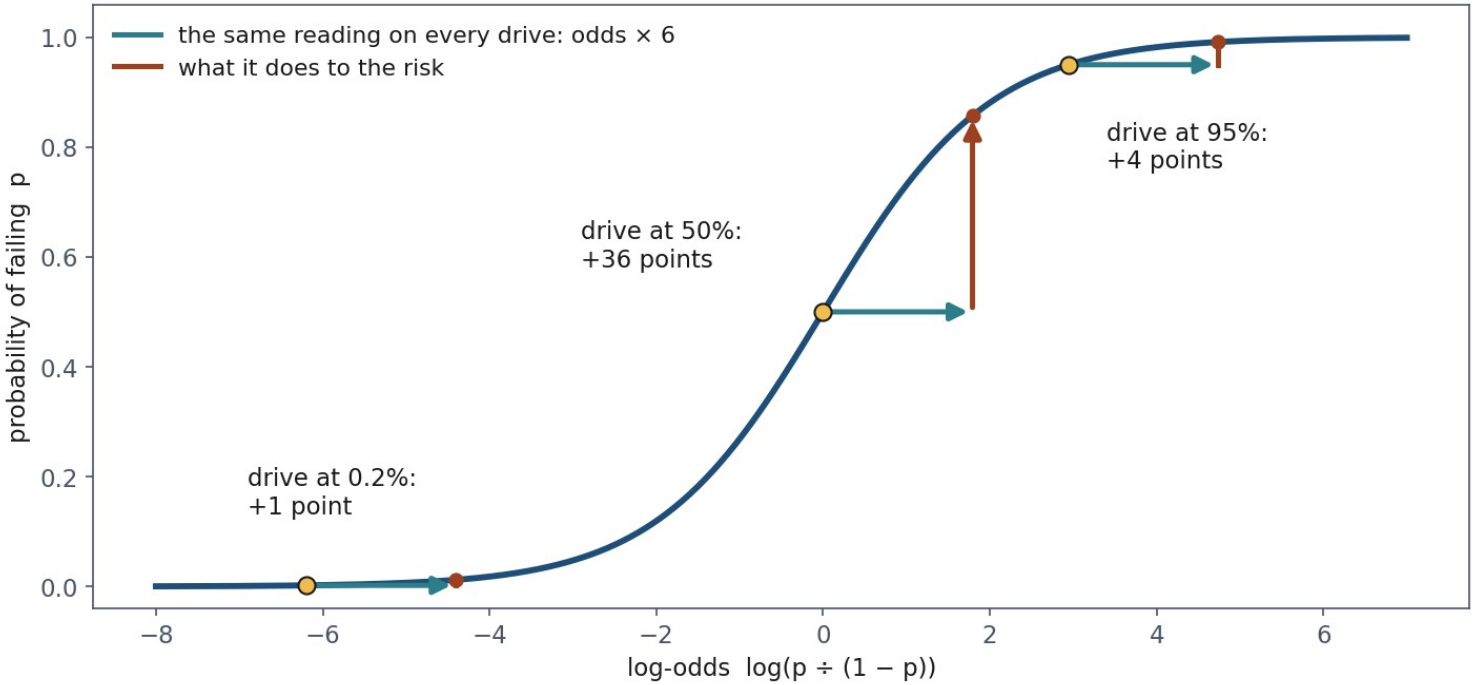
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Why the log-odds

- A line gives any number; a probability, only [0, 1]
- Odds, $p \div (1 - p)$: no ceiling, from 0 to ∞
- Log of the odds: no floor either, $-\infty$ to $+\infty$
- $p = 0.99$: $\log(0.99 \div 0.01) = \log 99 = +4.60$; $p = 0.5$: $\log(0.5 \div 0.5) = \log 1 = 0$
- Bonus: coefficients multiply the odds (slide 14)

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The same evidence on three drives



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Log-odds of p = P(y = 1 | x): a line

$$\log \frac{p}{1-p} = w^\top x + b \quad \Longrightarrow \quad p = \frac{1}{1 + e^{-(w^\top x + b)}}$$

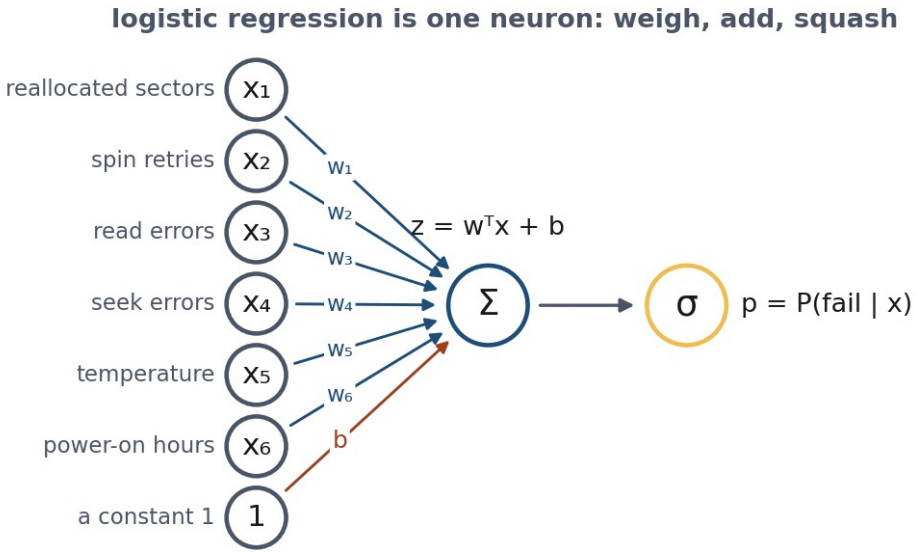
UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

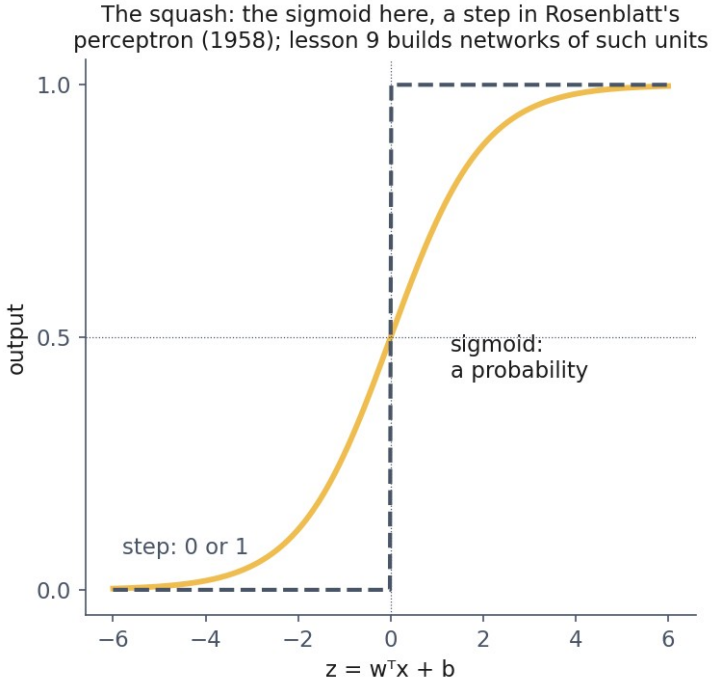
The model, in one line

$$P(y = 1 \mid x) = \sigma(w^\top x + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}$$

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

The model, drawn: one neuron





UNIVERSITÀ DEGLI STUDI DELL'AQUILA

What the coefficients mean

- Standardised (mean 0, SD 1): one SD more of reallocated sectors adds its true coefficient, 1.80 (slide 15), to the log-odds
- Adding to a log multiplies the **odds**: log(odds) + 1.80 = log(odds × 6.05), since exp(1.80) = 6.05
- So: **six times the odds** of failing, slide 10’s teal step
- In code: `make_pipeline(StandardScaler(), LogisticRegression())`

UNIVERSITÀ DEGLI STUDI DELL’AQUILA

The truth, and what we recovered

Feature	True	Fitted, notebook 1	Odds ×, exp(Fitted)
reallocated_sectors	1.80	1.66	5.28
spin_retry_count	1.15	1.17	3.22
read_error_rate	0.85	0.87	2.39
temperature_c	0.45	0.46	1.59
power_on_hours	0.35	0.38	1.46
seek_error_rate	**0.00**	**0.02**	**1.02**

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The intercept is the average drive’s chance

- Fitted intercept: **−6.09**, the log-odds of a drive average on every counter (all standardised x = 0)
- σ(−6.09) = 1 ÷ (1 + exp(6.09)) = 1 ÷ 442 = 0.0023: a 0.2% chance
- The fleet fails at 3.8% (slide 6): the failures come from the few drives far above average

Fitting means minimising something, but what?

- Squared error worked for lesson 3
- Why not just apply it to the sigmoid’s output?

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Step 1: each drive gets a p

Drive	What happened	y	Setting 1: $p = \sigma(w^T x + b)$, the P(fail)
A	failed	1	0.8
B	stayed healthy	0	0.1
C	stayed healthy	0	0.2

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Step 2: how likely was the outcome?

Drive	y	p	Probability of what happened: p if y = 1, 1 - p if y = 0
A	1	0.8	0.8
B	0	0.1	$1 - 0.1 = 0.9$
C	0	0.2	$1 - 0.2 = 0.8$
All three together, if the drives are independent: multiply, as for two coins (½ × ½ = ¼)			$0.8 \times 0.9 \times 0.8 =$ **0.576**

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Step 2 as formulas: one drive, then the likelihood

$$P(y \mid x) = p^{\,y}\,(1-p)^{1-y}$$

$$\mathcal{L}(w,b) = \prod_{i=1}^{m} \left(p^{(i)}\right)^{y^{(i)}} \left(1-p^{(i)}\right)^{1-y^{(i)}}$$

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Step 3: two settings, the likelier wins

Drive	y	Setting 1: p	Setting 1: probability of what happened	Setting 2: p	Setting 2: probability of what happened
A	1	0.8	0.8	0.3	0.3
B	0	0.1	$1 - 0.1 = 0.9$	0.6	$1 - 0.6 = 0.4$
C	0	0.2	$1 - 0.2 = 0.8$	0.5	$1 - 0.5 = 0.5$
Likelihood: the product			$0.8 \times 0.9 \times 0.8 =$ **0.576**		$0.3 \times 0.4 \times 0.5 =$ **0.060**

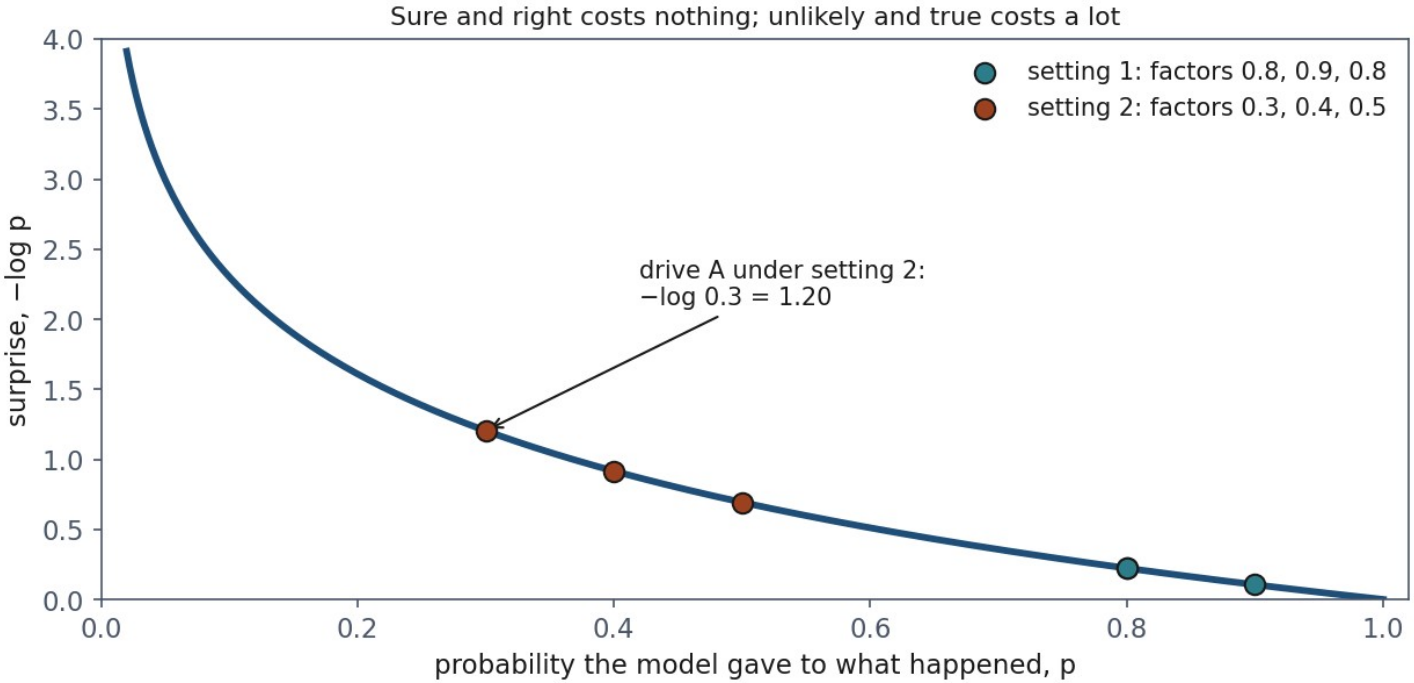
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Step 4: minus the log, a cost

Drive	y	Setting 1: −log of probability of what happened	Setting 2: −log of probability of what happened
A	1	−log 0.8 = 0.22	−log 0.3 = 1.20
B	0	−log 0.9 = 0.11	−log 0.4 = 0.92
C	0	−log 0.8 = 0.22	−log 0.5 = 0.69
Sum = −log of the likelihood		0.22 + 0.11 + 0.22 = **0.55**	1.20 + 0.92 + 0.69 = **2.81**

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Surprise: minus the log of a probability



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Cross-entropy, also called log loss

$$L = -[\, y \log p + (1-y) \log(1-p)]$$

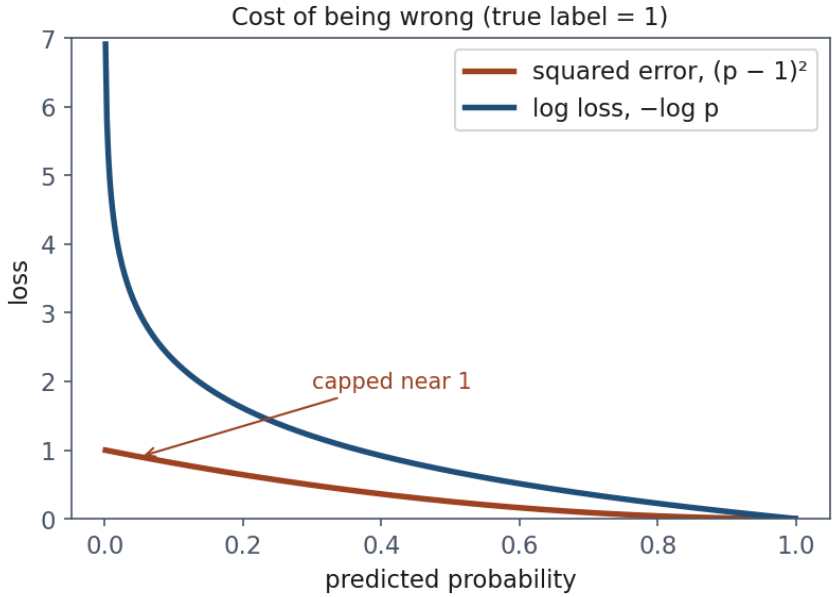
UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

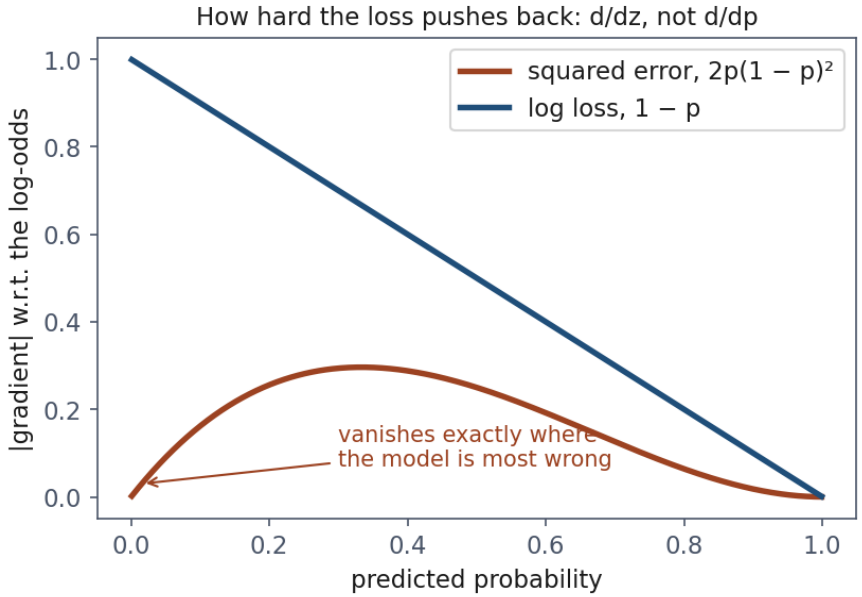
Lesson 3’s gradient, $\hat{y} = \sigma(w^T x + b)$

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^{m} (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

So why not squared error?





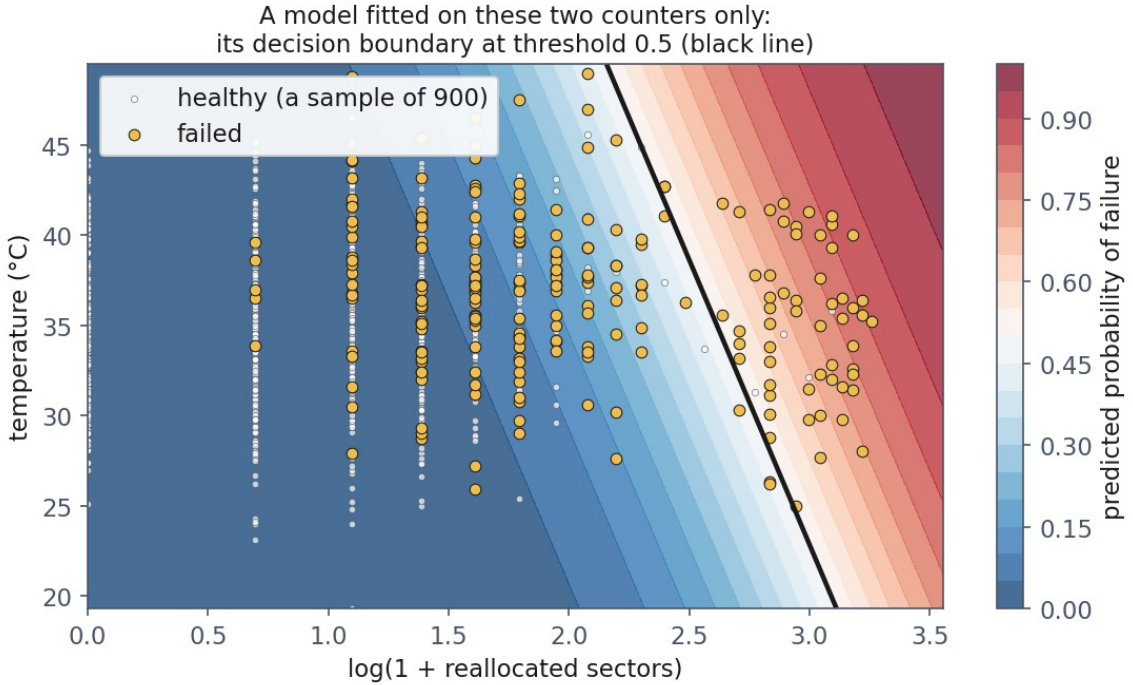
UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

And it is convex, which squared error is not

- Log loss is **convex** in the weights, so any local minimum is global
- Squared error through a sigmoid is not
- Why not, from slide 26: its slope falls from 0, then climbs back to 0; a convex curve’s slope only grows

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The decision boundary of a two-counter model



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Notebook 1, live

section, and its slides	you call (→ the number to come back with)
a line on a 0/1 label, 7	`LinearRegression` → 3,606 negative probabilities
odds, and why not squared error, 8-27	NumPy only: the sigmoid, the two losses, their gradients
gradient descent, 25	`train_test_split`, `StandardScaler`, fifteen lines of NumPy → log loss 0.6931 to 0.0692
the decision boundary, 28	`make_pipeline(StandardScaler(), LogisticRegression())` on two features
the truth, recovered, 15-16	`LogisticRegression`, `.coef_`, `.intercept_` → seek_error_rate 0.02, intercept −6.09

Break

- Ten minutes

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Two models

Model, on the 2,000 test drives (76 fail)	Accuracy = right ÷ 2,000	Failures caught
Predict “healthy” for everything	every healthy drive: 1,924 ÷ 2,000 = **96.20%**	**0 of 76**
Logistic regression, threshold 0.5	43 failures + 1,911 healthy: 1,954 ÷ 2,000 = 97.70%	43 of 76

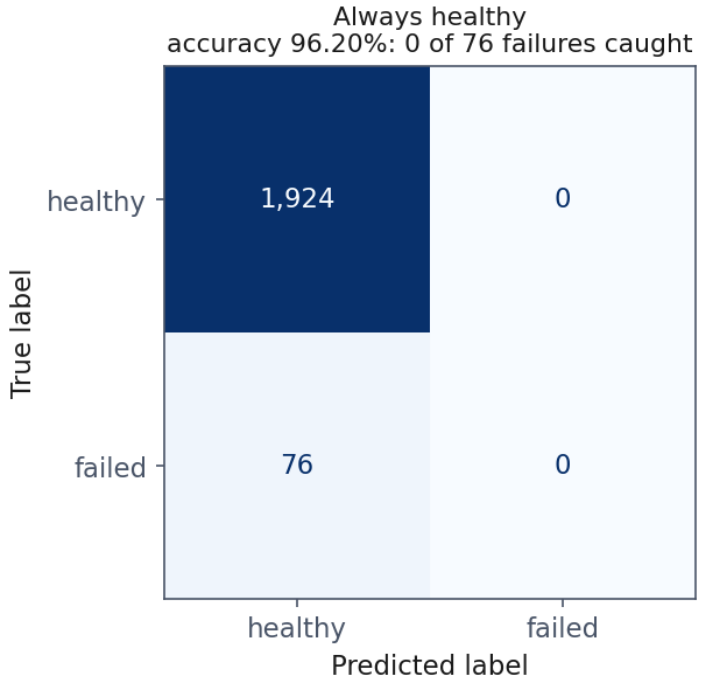
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

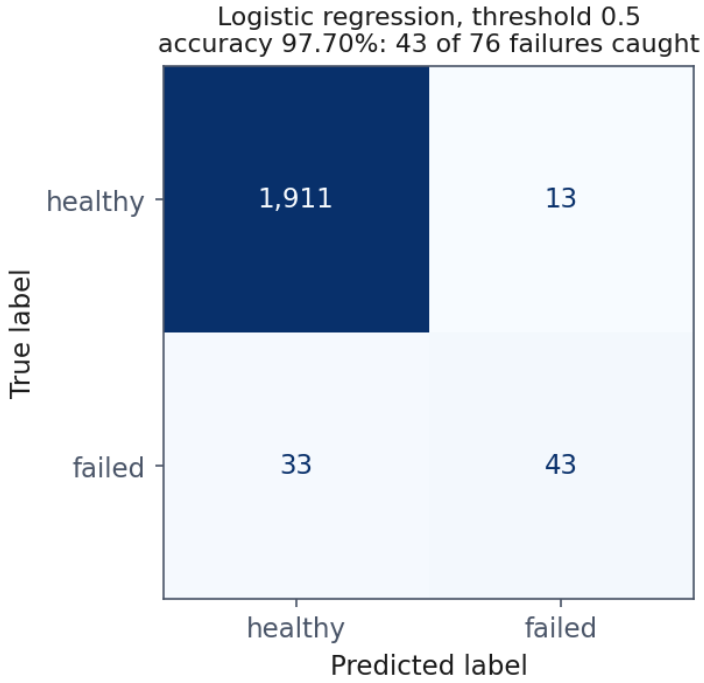
Why everyone makes this mistake once

- Accuracy is the fraction of predictions that were right
- On a **balanced** problem whose two errors cost the same, that is exactly what you want
- The interesting class is almost always the **rare** one

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Four numbers instead of one





UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The four outcomes, and what they cost

- **False positive**: we replace a healthy drive. A technician’s hour, €140
- **False negative**: a drive dies in service. Callout, rebuild, risk, €2,600
- Both costs are assumptions supplied from outside the data, not measured
- A correct decision counts as €0: each cost is what the error adds over the right call

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Precision and recall: look at the denominator

Precision = TP / (TP + FP)
"of the alarms we raised"

	predicted healthy	predicted failing
actually healthy	TN 1911	**FP** 13
actually failed	FN 33	**TP** 43

Recall = TP / (TP + FN)
"of the drives that failed"

	predicted healthy	predicted failing
actually healthy	TN 1911	FP 13
actually failed	**FN** 33	**TP** 43

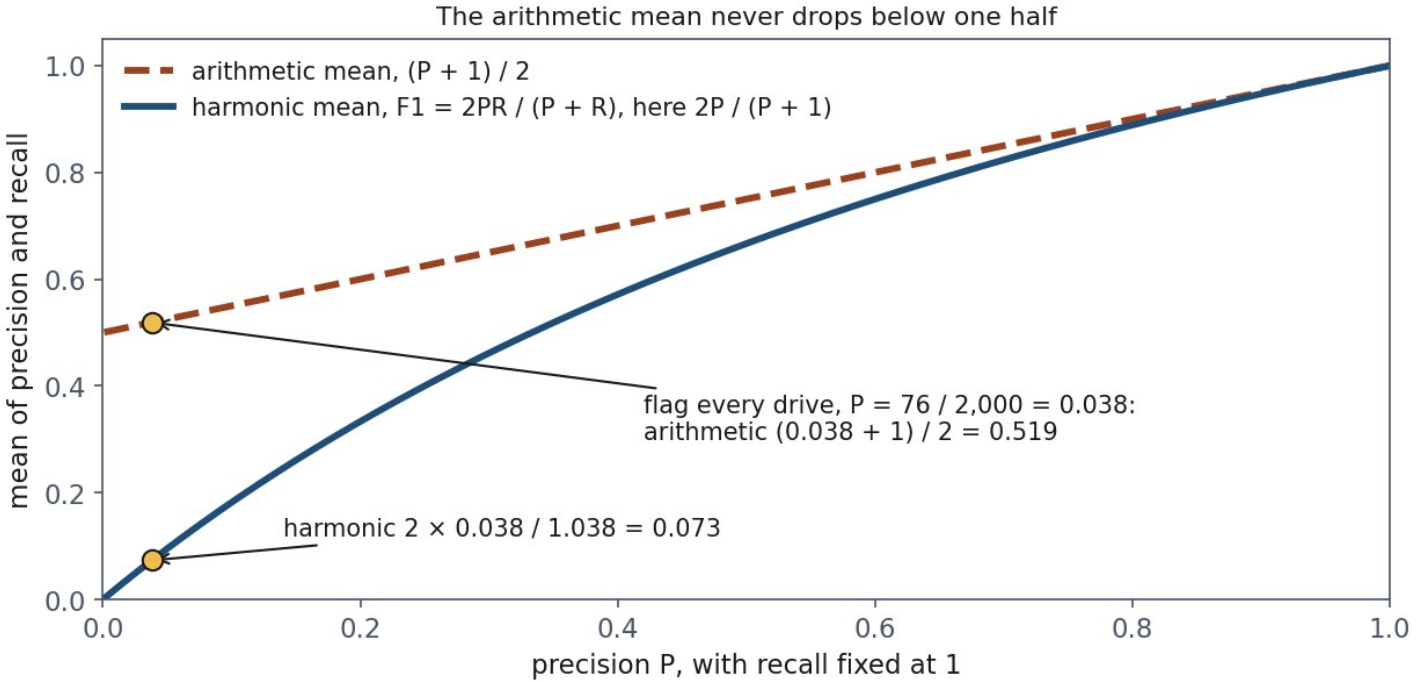
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Our model, in those terms

- True and false positives and negatives: TP, FP, FN, TN
- Precision = TP ÷ (TP + FP) = 43 ÷ (43 + 13) = **0.768**: of the drives flagged
- Recall = TP ÷ (TP + FN) = 43 ÷ (43 + 33) = **0.566**: of the drives that failed
- Specificity = TN ÷ (TN + FP) = 1,911 ÷ 1,924 = 0.993; 1 − 0.993 is the false positive rate (FPR)
- In code: `precision_score`, `recall_score`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

F1: why the harmonic mean



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

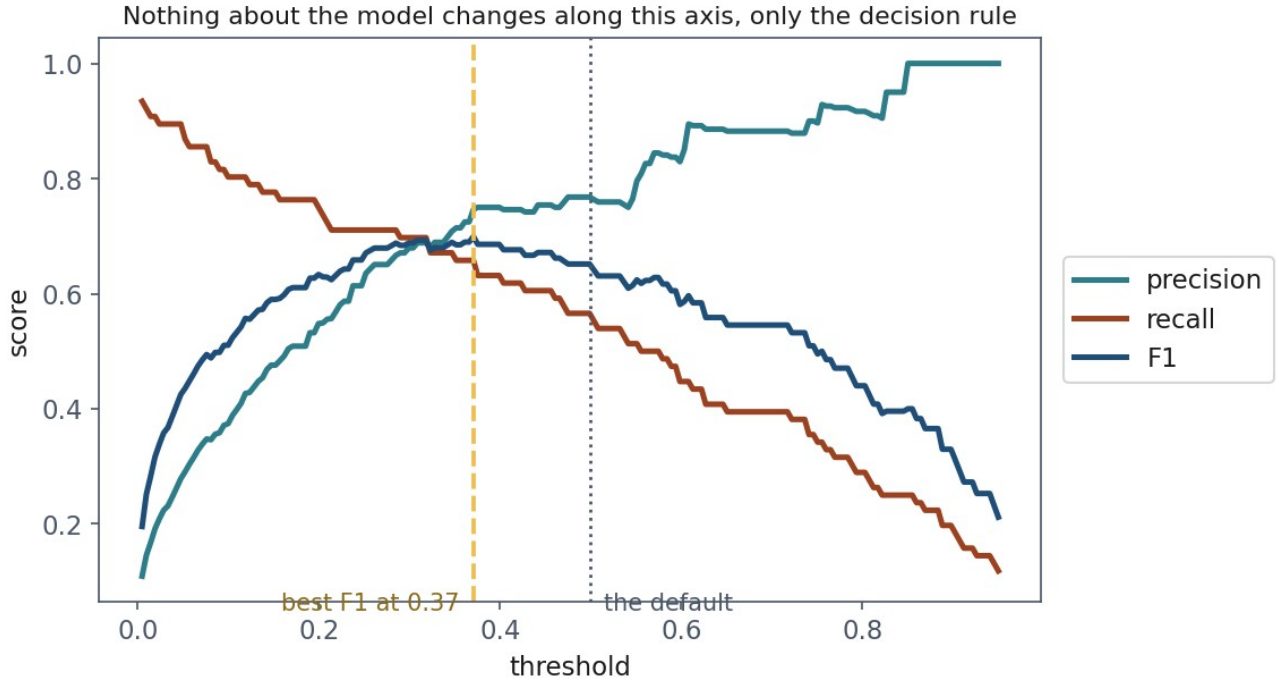
Read the minority class, not the average

- F1 per class: healthy 0.988 (1,924 drives), failed 0.652 (76 drives)
- `macro avg` F1 = (0.988 + 0.652) ÷ 2 = **0.820**: both classes count the same
- `weighted avg` F1 = (1,924 × 0.988 + 76 × 0.652) ÷ 2,000 = **0.975**: the healthy class dominates
- In code: `classification_report(y_test, y_pred)` prints both rows

Notebook 2, live

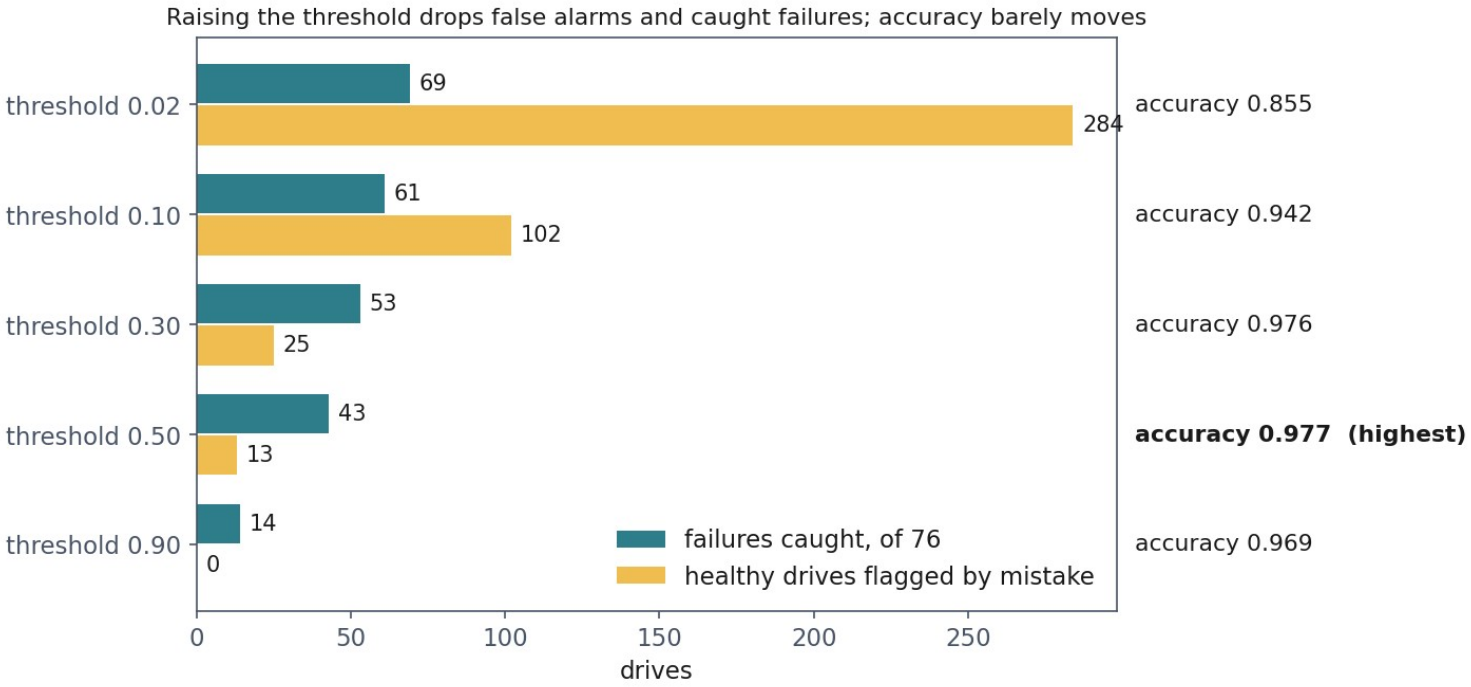
section, and its slides	you call (→ the number to come back with)
the number that should worry you, 31	`accuracy_score` → 96.20% for “always healthy”
the confusion matrix, 33	counts by hand, then `confusion_matrix`, `ConfusionMatrixDisplay` → TP 43
precision and recall, 35-38	`precision_score`, `recall_score`, `f1_score`, `classification_report`
the threshold sweep, 40-41	`predict_proba`, then the metrics at each threshold
the precision-recall curve, 42	`precision_recall_curve`, `average_precision_score` → 0.717 against 0.038

The threshold is a choice nobody made



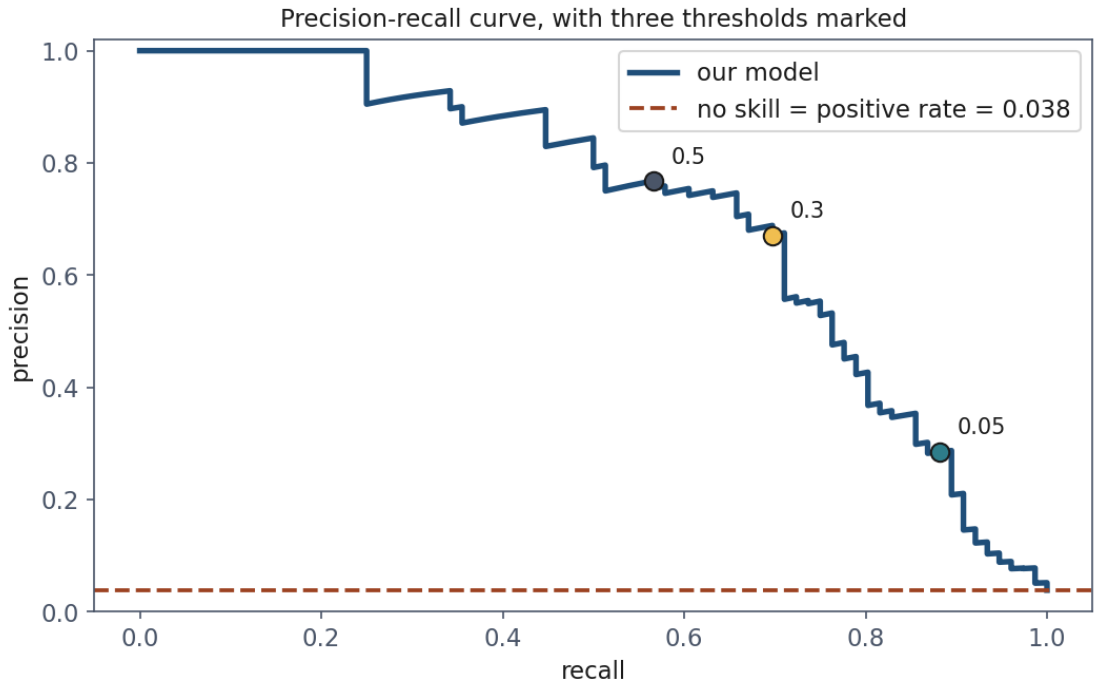
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

What the sweep actually costs



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The precision-recall curve

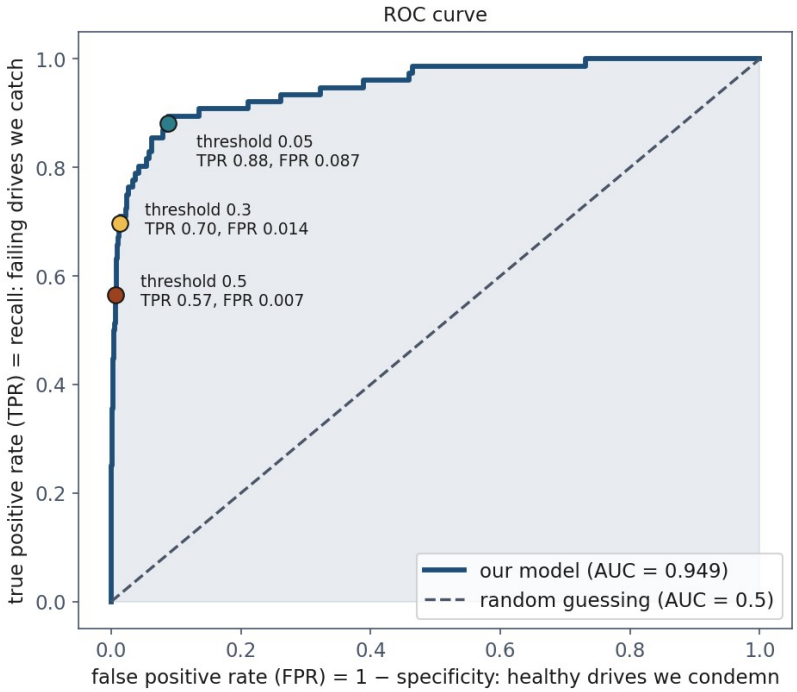


UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Comparing models without choosing a threshold

- We keep evaluating at one threshold, but which?
- The **ranking**: the 2,000 test drives sorted by predicted probability, riskiest first
- A threshold is a cut in this list: every drive above the cut is flagged
- What if we score the whole ranking, every cut at once?

The ROC curve



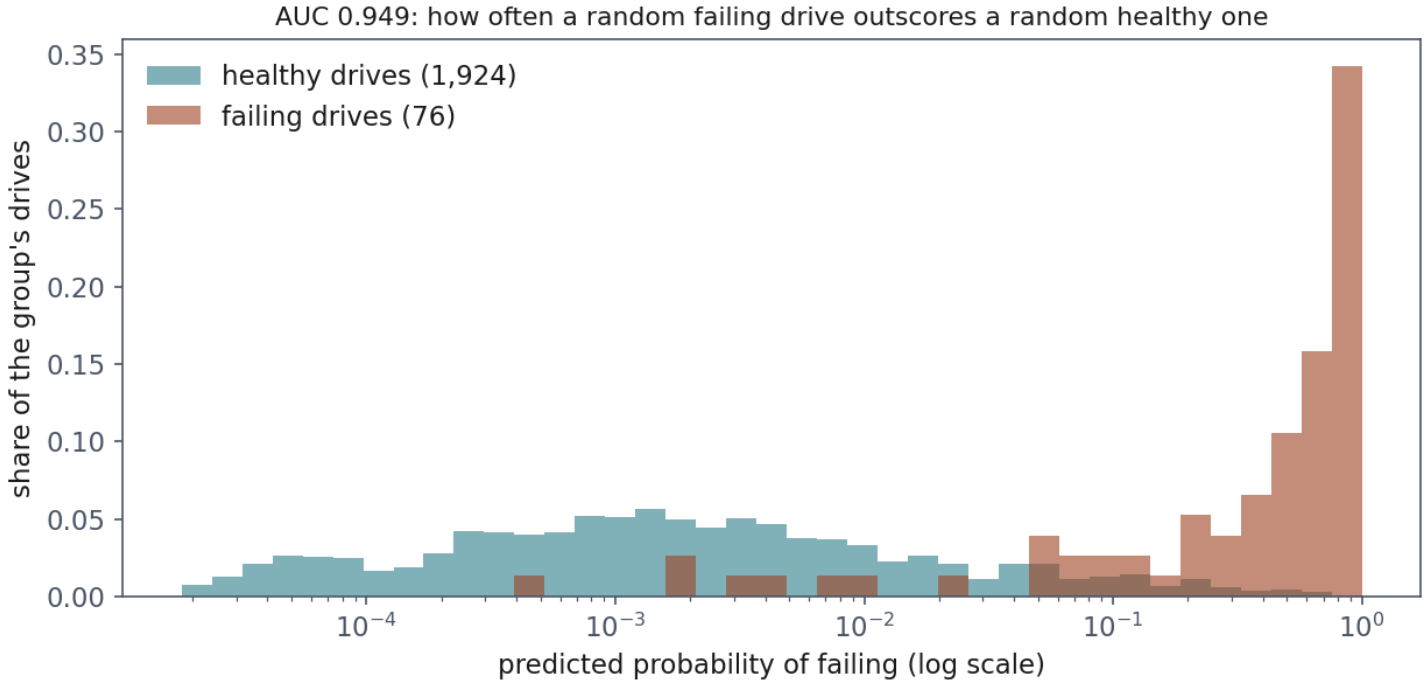
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

What the area means

- **Area under the curve (AUC) = the probability that a random failing drive scores above a random healthy one**
- Ours: 0.949 (`roc_auc_score`, notebook 3)
- Check: in 200,000 random pairs, the failing drive scores higher 94.9% of the time
- In code: `roc_auc_score(y_test, scores)`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The scores behind the AUC



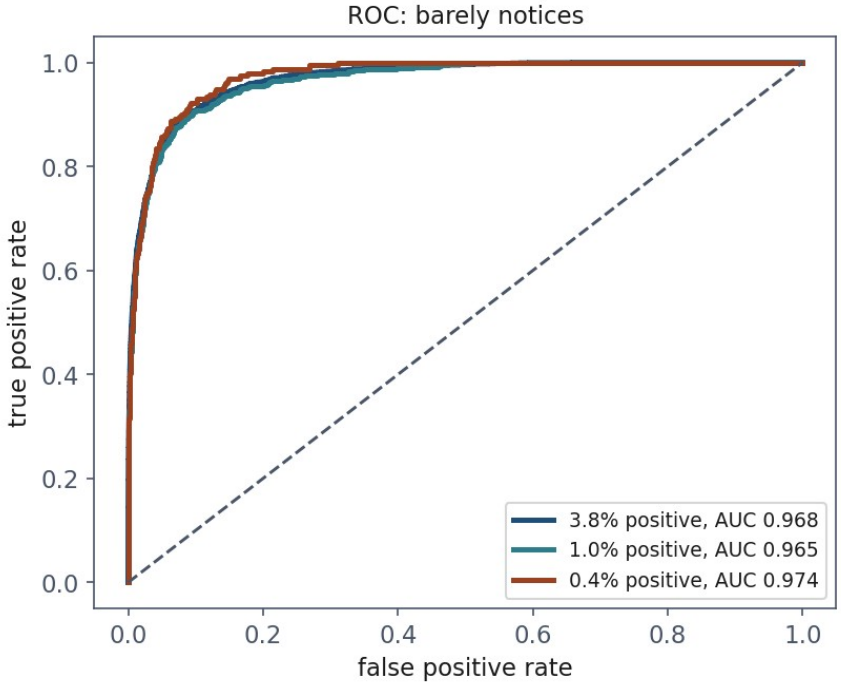
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

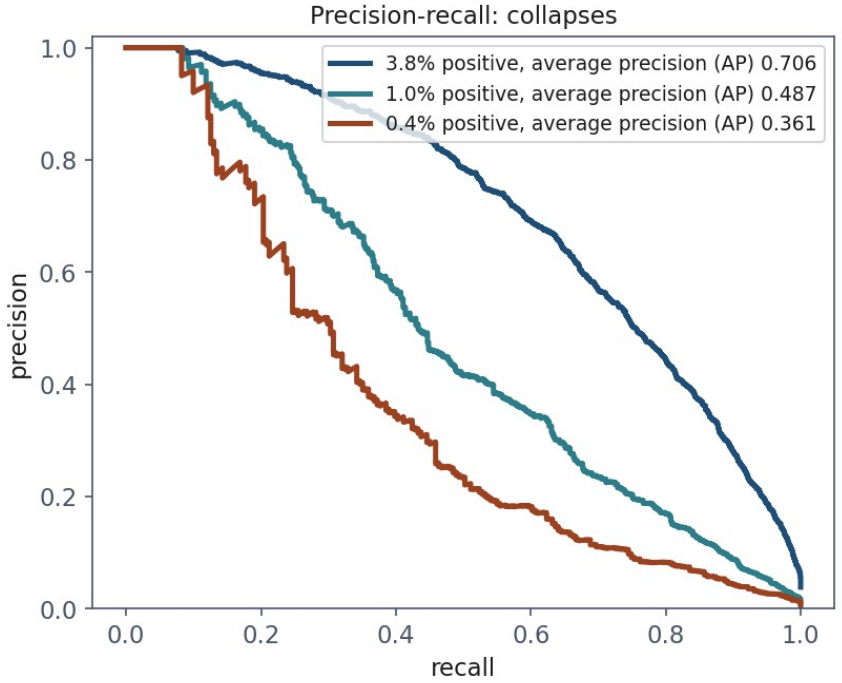
What AUC does not tell you

- Divide every predicted probability by 10
- AUC is **unchanged**, 0.949: the order of the drives is the same
- Every decision changes: at threshold 0.5, 56 drives flagged before, none after (notebook 3)
- The model is now badly **calibrated**: calibrated means a predicted 0.7 comes true about 70% of the time

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Where ROC misleads





UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Why the two disagree

- False positive rate (FPR) = FP ÷ (FP + TN) divides by **all 57,575 healthy drives**, untouched by the thinning: 304 ÷ 57,575 = 0.0053 in all three fleets
- Precision = TP ÷ (TP + FP): FP **stays 304** while TP falls from 1,127 to 106, so the false alarms' share of the alarms grows from 0.21 to 0.74

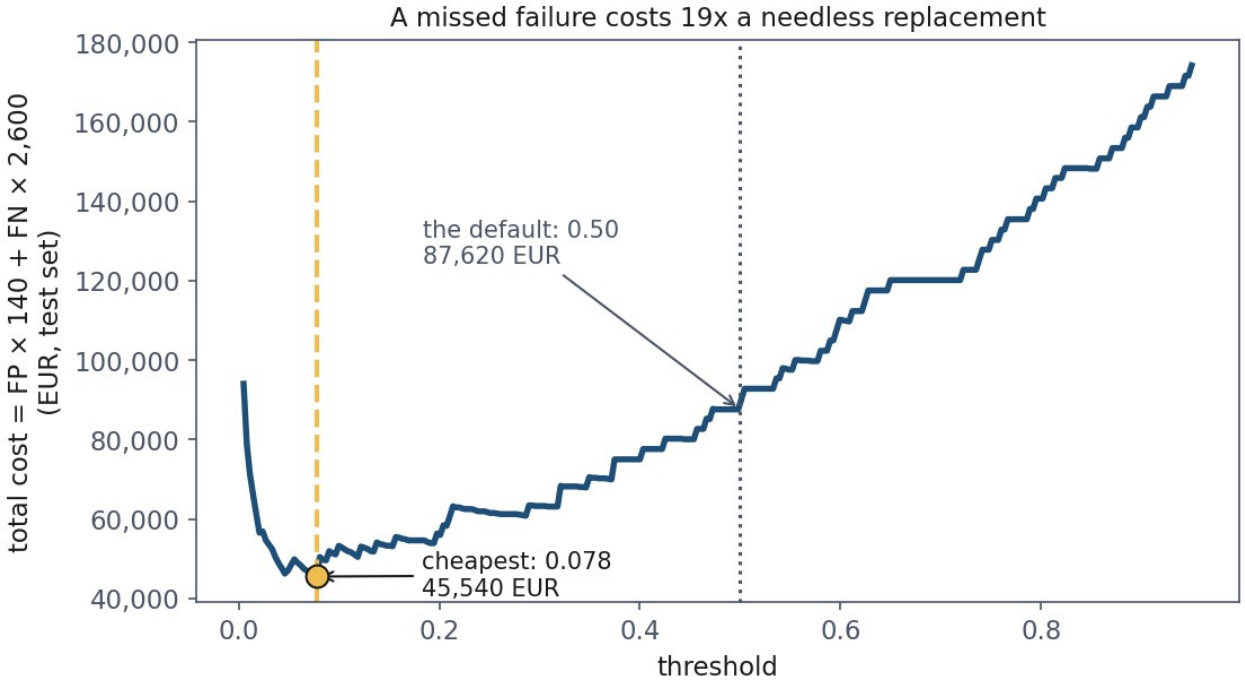
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The rule: report both

- AUC: are failures ranked above healthy drives? Precision-recall: how many alarms are real?
- On imbalanced data report the **precision-recall curve beside the AUC**, not instead
- **Average precision (AP)**: the area under that curve, $\Sigma_k (R_k - R_{k-1}) \times P_k$, each threshold's precision weighted by the recall it adds
- Ours: 0.717 (notebook 2); a random ranking: 76 ÷ 2,000 = 0.038

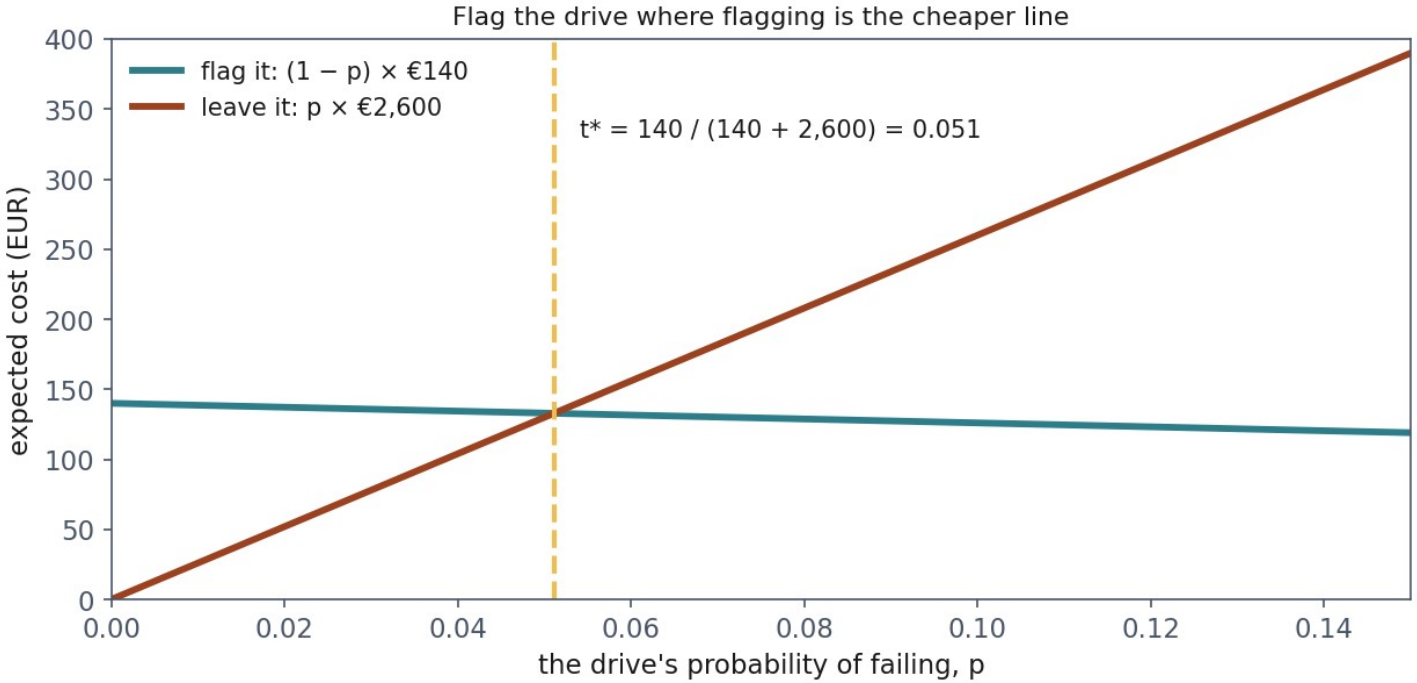
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Choosing the threshold from money



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Where the two costs cross



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Where they cross, in one line

flag if $p\, C_{FN} > (1 - p)\, C_{FP}$

$$\Leftrightarrow \; p > \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{140}{2740} = 0.051 = t^{*}$$

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The caveat you must state

- We chose the threshold **by looking at the test set**
- That is precisely what lesson 1 forbade
- Done properly: choose it on a **validation set**, then measure once on the test set (lesson 5)
- The price of looking, here: t* = 0.051, fixed before looking, costs €49,100 on the test set; 0.078, picked by looking, €45,540

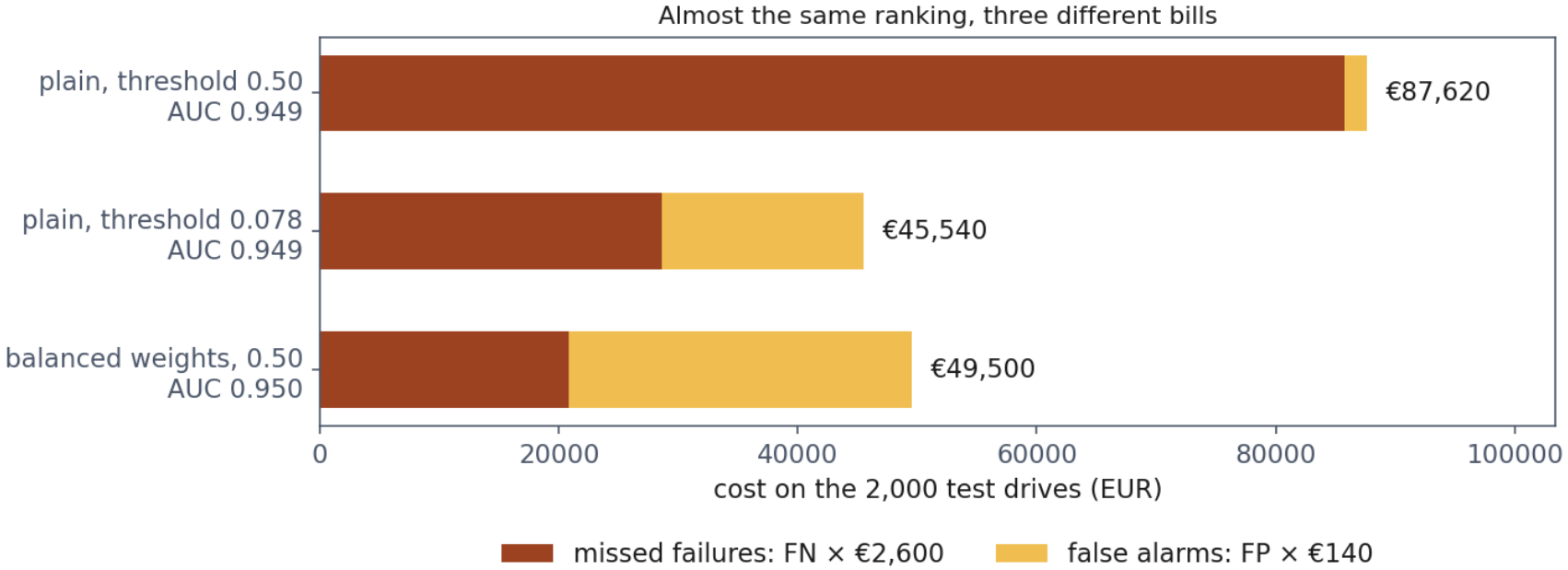
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

What class weights do

- Training multiplies each drive’s log-loss term by its **class weight**
- `class_weight="balanced"`: m ÷ (2 × class size), 0.52 healthy, 13.0 failed
- A failure counts as 25 healthy drives: **class frequencies**, not costs
- Same decisions as the plain model at 1 ÷ 26 = 0.038, on 1,987 of 2,000 drives

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Class weights: the same lever, pulled earlier



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Notebook 3, live

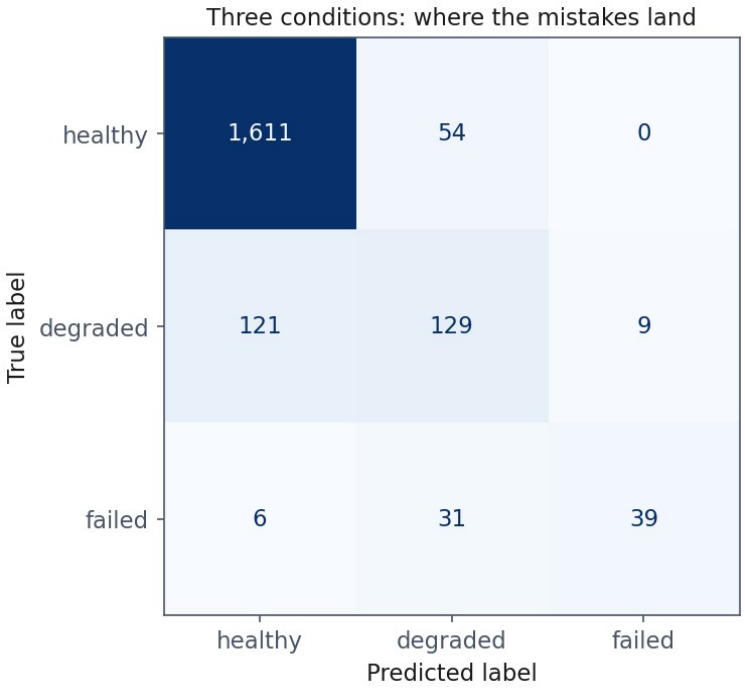
section, and its slides	you call (→ the number to come back with)
the ROC curve by hand, 44	`roc_curve`, `roc_auc_score` → 0.9493
what AUC means, 45-46	200,000 random (failing, healthy) pairs → 0.9488
the rare-positive trap, 48-49	`average_precision_score` beside `roc_auc_score` → AP 0.706 to 0.361, AUC unmoved
the threshold from costs, 51-54	`confusion_matrix` at each threshold → 45,540 EUR at 0.078, against 87,620 at 0.50
class weights, 55-56	`LogisticRegression` with balanced class weights → AUC 0.950
three classes, 58-60	`classification_report` → macro F1 0.707, weighted 0.883

Three classes: softmax

- One score per class, $z_k = w_k^T x + b_k$; **softmax**: exponentiate, divide by the sum
- A degraded drive: scores 0.615, 0.603, −1.218 → 0.466, 0.460, 0.074, called healthy
- Predicted class = the largest probability, no 0.5 threshold; two classes: the sigmoid of $z_1 - z_0$
- In code: `decision_function` (scores), `predict_proba` (softmax)

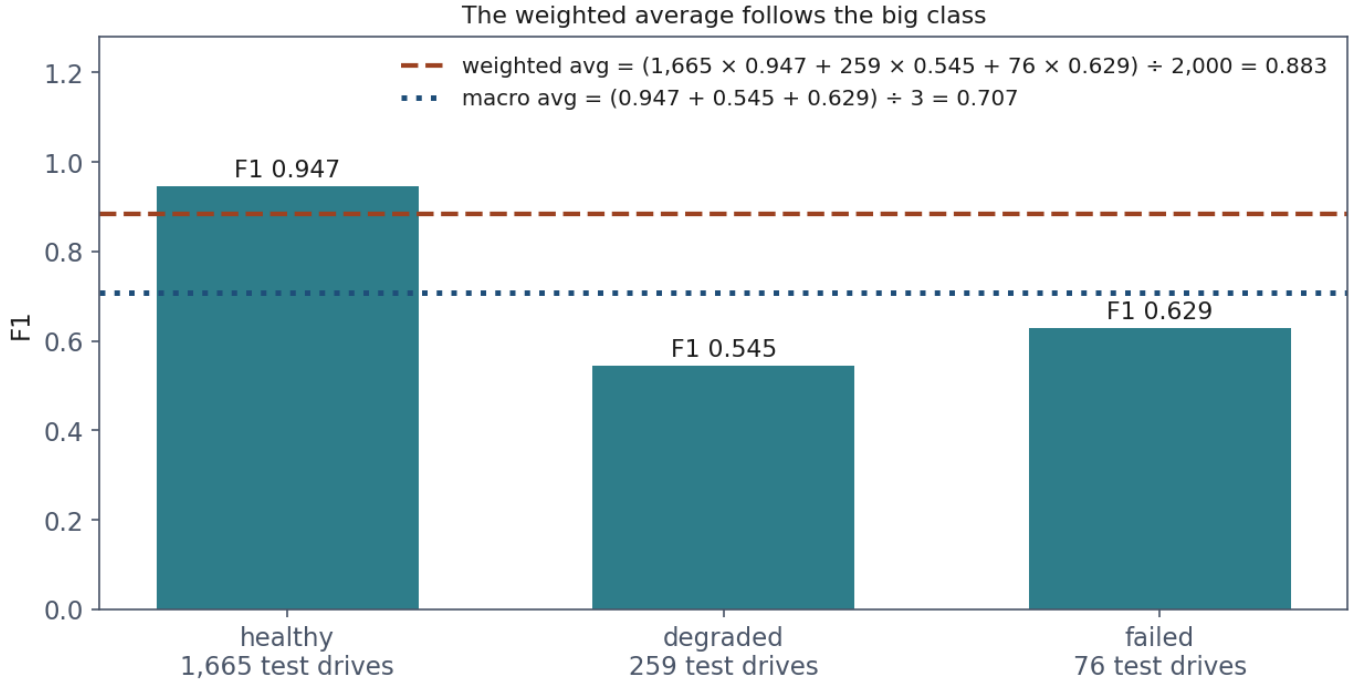
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

More than two classes



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Macro against weighted, one last time



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

What to take away

- Logistic regression: sigmoid of a linear score, fitted by log loss
- **96.20% accuracy, nothing caught**: check the baseline, 1,924 ÷ 2,000
- Precision: the **alarms**; recall: the **failures**
- **Threshold** from costs: 140 ÷ 2,740 = 0.051, if calibrated
- **AUC = ranking**: blind to calibration, hides imbalance

Homework

- **Exercise 4**, discussed at the start of **Friday 6 November**
- Build a classifier, choose a threshold from a stated cost ratio, defend it

UNIVERSITÀ DEGLI STUDI DELL'AQUILA